Group II, Row IV (final-state $g \to gg$): result rewritten to match the note *"Divergence structure of the final-state $g \to gg$ contribution"*

Conventions

$d_\perp = d = 2 - \epsilon$ (our ϵ); the note's $\epsilon_{\rm note}$ has $d = 2 - 2\epsilon_{\rm note}$, so $\frac{2}{\epsilon} = \frac{1}{\epsilon_{\rm note}}$. $\mu \equiv \mu_{\overline{MS}}$. Collinear poles are IR ($\epsilon < 0$). Accuracy: $O(\Lambda/k^+,\, k^+/\vee)$ dropped, as in the note, Eq. (64). The note's prefactor is $\bar{C}/k^+ = \frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^5}\frac{1}{k^+}$.

Result

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{d^3N_{\rm DGLAP}}{d^3k} + \frac{d^3N_{\rm RC}}{d^3k} + \frac{d^3N_{\rm finite}}{d^3k} + \frac{d^3N_{\rm BFKL}}{d^3k} + \frac{d^3N_{\rm JIMWLK}}{d^3k}\,, \tag{1}$$

$$\begin{aligned}\frac{d^3N_{\rm DGLAP}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\frac{2}{\epsilon}\int_0^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\\ &\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,,\end{aligned} \tag{2}$$

$$\mathcal{P}_{gg}(\xi) = 2N_c\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big]+\frac{\beta_0}{2}\,\delta(1-\xi)\,,\qquad \beta_0=\frac{11N_c}{3}\,,$$

$$\int_0^1 d\xi\,\frac{\xi}{(1-\xi)_+}\,h(\xi) \equiv \int_0^1 d\xi\,\frac{\xi\,h(\xi)-h(1)}{1-\xi}\,,$$

$$\begin{aligned}\frac{d^3N_{\rm RC}}{d^3k} =& +\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\beta_0}{16\pi^2}\frac{1}{k^+}\frac{2}{\epsilon}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\\ &\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,,\end{aligned} \tag{3}$$

$$\begin{aligned}\frac{d^3N_{\rm finite}}{d^3k} =& -\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\left\{\frac{\xi\,\mathcal{G}(\xi)-\mathcal{G}(1)}{1-\xi}+\frac{(1-\xi)\,\mathcal{G}(\xi)-\mathcal{G}(0^+)}{\xi}+\xi(1-\xi)\,\mathcal{G}(\xi)\right\}\\ &+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\ &\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,,\end{aligned} \tag{4}$$

with

$$\begin{aligned}\mathcal{G}(\xi) \equiv& \frac{1}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4\,\xi^2}\Big]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\\ &\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,,\end{aligned} \tag{5}$$

$$\mathcal{G}(1) = \int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\Big[2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,, \tag{6}$$

$$\mathcal{G}(0^+) = -\frac{4\pi}{k^2}\int_{x,x',z}\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,, \tag{7}$$

$$\frac{d^3N_{\rm BFKL}}{d^3k} = +\,\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,z,w,w',x'} e^{-ik(w'-w)}\,\frac{(w-z)\cdot(w'-z)\,(x-z)\cdot(x'-z)}{(w-z)^2(x-z)^2(w'-z)^2(x'-z)^2}$$
$$\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\times\,[U^{ce}(z)-U^{ce}(x)]\;\rho^{e'}(x')\rho^{e}(x)\,, \tag{8}$$

$$\frac{d^3N_{\rm JIMWLK}}{d^3k} = +\,\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,z,w,w',x'} e^{-ik(w'-w)}\,\frac{(w-z)\cdot(w'-z)\,(x-w)\cdot(x'-w')}{(w-z)^2(x-w)^2(w'-z)^2(x'-w')^2}$$
$$\Big[U^{ce'}(w')-U^{ce'}(x')\Big]\times\,[U^{ce}(w)-U^{ce}(x)]\;\rho^{e'}(x')\rho^{e}(x)$$
$$=-\,\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\,, \tag{9}$$

where in the JIMWLK term the z integral is done in $d=2-\epsilon$ (it is log-divergent at large $|z|$ in $d=2$).

Map to the note

Here	Note	
DGLAP	$-\frac{\pi}{\epsilon}\int_0^1\frac{d\xi}{\xi^2}\big[\hat P_+ + \frac{11}{12}\delta(1-\xi)\big]\mathcal{L}(k/\xi)$	Eq. (65), "full P_{gg}: $\overline{\rm MS}$ FF counterterm"
RC	$+\frac{\pi}{\epsilon}\frac{11}{12}\mathcal{L}(k)$	Eq. (65), "cancels α_s counterterm"
JIMWLK	$\ln\frac{k^+}{\Lambda}\big[\mathcal{J}_{\rm fin}-\frac{\pi}{\epsilon}\mathcal{L}(k)\big]$	Eq. (64)
BFKL	$\ln\frac{\vee}{k^+}\,\mathcal{B}(k)$	Eq. (64)
finite	$F(k;\mu)=\int_0^1 d\xi\,[I_{\rm fin}-\mathcal{J}_{\rm fin}/(1-\xi)-\mathcal{B}/\xi]$	Eq. (34)

Dictionary: $-\pi\,\mathcal{G}(\xi)$ is the finite part of the note's $G(\xi)$, Eq. (51); $\mathcal{J}_{\rm fin}=-\pi\,\mathcal{G}(1)$, $\mathcal{B}=-\pi\,\mathcal{G}(0^+)$; the bracket term is the traceless line of Eq. (50). The braces in the finite term are $\hat P(\xi)\mathcal{G}(\xi)-\mathcal{G}(1)/(1-\xi)-\mathcal{G}(0^+)/\xi$ with $\hat P=\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}+\xi(1-\xi)$; each fraction is integrable at its endpoint.

What was changed relative to the previous version

1. $\log\frac{(y'-y)^2\mu^2}{4}\to\log\frac{(y'-y)^2\mu^2}{4\xi^2}$ at $\xi\neq1$ (the note continues the relative momentum ℓ of the pair to d dimensions, Eq. (50)). This is the only change that alters the sum, by $+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_0^1 d\xi\,\hat P(\xi)\frac{\log\xi^2}{\xi^2}\int e^{-ik\cdot(y'-y)/\xi}(\dots)$.
2. DGLAP and RC now contain only the poles; their finite parts moved into "finite". The finite parts of the $\frac{\beta_0}{2}\delta(1-\xi)$ term and of the old RC term were equal and opposite, so nothing is lost.
3. $k^+/\vee\to0$ in the lower limits. The old "$(1-\xi)/\xi$" subtraction line and the boundary term $\propto[\frac{1}{\vee}-\frac{1}{k^+}]$ combine exactly into the single $-\mathcal{G}(0^+)/\xi$ subtraction inside "finite" (old S line + boundary $=-\frac{\bar C}{k^+}\mathcal{B}\int_{k^+/\vee}^1\frac{d\xi}{\xi}$).
4. The old $\int_{k^+/\vee}^1 d\xi\,\frac{\xi}{(1-\xi)_+}$ becomes the $\int_0^1$ plus distribution of the note, Eq. (33); the difference is $O(k^+/\vee)$.
5. BFKL and JIMWLK are unchanged.

Remarks (as in the note)

Net-colour IR pole. With $Q^c\equiv\int_x[\delta^{ce}-U^{ce}(x)]\rho^e(x)\neq0$, the JIMWLK, BFKL and finite terms contain the large-$|y|$ pole

$$-\frac{1}{(2\pi)^3}\frac{g^4N_c}{2\pi^2}\frac{1}{k^+k^2}\,\frac{2}{\epsilon}\,Q^cQ^c\Big[\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}-\frac{11}{6}\Big], \tag{10}$$

which is the note's Eq. (63). It can be pulled out explicitly as in Eq. (65); it vanishes iff $Q^c=0$.

After counterterms (note, Sec. 7.5). The $\overline{\rm MS}$ fragmentation counterterm cancels the DGLAP pole and the coupling counterterm cancels the RC pole. The $\frac{11}{12}$ share of the $\log(y'-y)^2$ in $\mathcal{G}(1)$ then sets the scale of one power of α_s inside the LO integral, $\alpha_s\big(4e^{-2\gamma_E}/|y-y'|^2\big)$ (note, Eq. (55)).